

- 5 Phloem and xylem are specialised tissues involved in the transport of water, ions and assimilates in plants.

(a) A photomicrograph of a transverse section of a dicotyledonous stem is shown in Fig. 5.1.

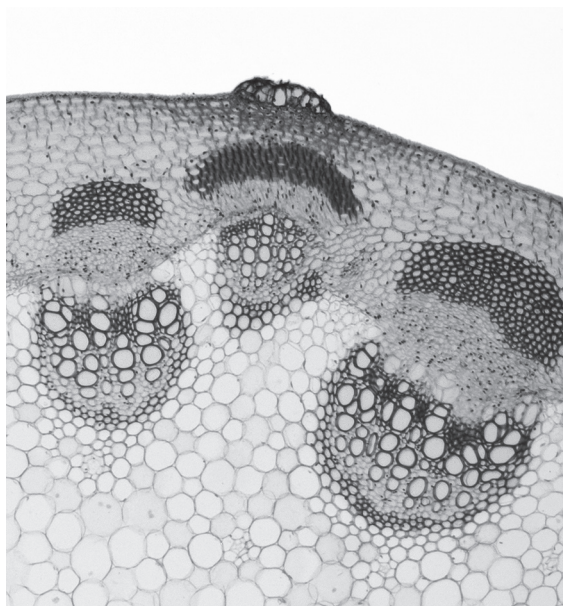


Fig. 5.1

Use label lines and labels to identify the phloem tissue **and** xylem tissue in Fig. 5.1. [2]

- (b) (i) Describe how sucrose is transported in phloem sieve tubes from photosynthesising leaves to other parts of the plant.

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(ii) Cyanide ions (CN^-) inhibit the activity of an enzyme involved in respiration.

Suggest why the treatment of photosynthesising leaves with CN^- results in less sucrose being transported into phloem sieve tubes.

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(c) A student was asked to carry out semi-quantitative Benedict's tests on two solutions.

- Solution **A** was extracted from the cytoplasm of cells in the mesophyll tissue of photosynthesising leaves.
- Solution **B** was extracted from the phloem sap in phloem sieve tubes.

The solutions were taken from the same plant, and other variables were standardised.

For each solution, the student measured the time taken for the first colour change to appear.

Suggest which of the two solutions, **A** or **B**, would change colour in the shortest time.

Explain your answer.

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[Total: 10]